

```

>>> C=np.array([[1,2,3],[0,0,0],[0,0,0]])
>>> A-C
array([[0, 0, 0],
       [4, 5, 6],
       [7, 8, 8]])
>>> np.subtract(A,C)
array([[0, 0, 0],
       [4, 5, 6],
       [7, 8, 8]])
>>> A@C
array([[ 1,  2,  3],
       [ 4,  8, 12],
       [ 7, 14, 21]])
>>> np.matmul(A,C)
array([[ 1,  2,  3],
       [ 4,  8, 12],
       [ 7, 14, 21]])
>>> A*C
array([[1, 4, 9],
       [0, 0, 0],
       [0, 0, 0]])

```

Figure 6: More matrix operations

which among the eight matrices can be multiplied with each other. Though by the strict mathematical definition, a matrix is two-dimensional, we will be dealing with arrays of higher dimensions. So, just to give an example, let us see how a three-dimensional array is represented using NumPy. The following line of code creates a three-dimensional array named T.

```
T = np.array([[[11,22],[33,44]],[[55,66],[77,88]]])
```

An introduction to Pandas

So far, we were entering matrices from the keyboard. What if we want to read a large matrix from a text file or a data set (very large for that matter) and process it? Here comes another powerful Python library to our rescue, Pandas. Let us begin by reading a small CSV (comma-separated value) file, which is a type of text file (don't worry, soon we will read Microsoft Excel and LibreOffice/OpenOffice Calc files). Figure 8 shows how a CSV file called *cricket.csv* is read, and the first three lines in it are printed on the terminal. More features of Pandas will be discussed in the coming articles in this series.

Rank of a matrix

The rank of a matrix is the *dimension* of the *vector space spanned* by its rows (columns). The three italicised words,

```

>>> E@A
array([[30, 36, 39],
       [66, 81, 90]])
>>> A@E
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
ValueError: matmul: Input operand 1 has a mismatch in its core
dimension 0, with gufunc signature (n?,k),(k,m?)->(n?,m?) (si
ze 2 is different from 3)
>>> D@G
array([[1694]])
>>> G@D
array([[ 121,  242,  363],
       [ 242,  484,  726],
       [ 363,  726, 1089]])

```

Figure 7: More examples of matrix multiplication

```

>>> import pandas as pd
>>> dataframe = pd.read_csv('cricket.csv')
>>> dataframe.head(3)
   Name  Centuries
0  Sachin         49
1  Kohli         43
2  Ponting        30

```

Figure 8: Reading a CSV file with Pandas

```

>>> np.linalg.matrix_rank(A)
3
>>> np.linalg.matrix_rank(B)
1
>>> np.linalg.matrix_rank(C)
1
>>> np.linalg.matrix_rank(D)
1
>>> np.linalg.matrix_rank(E)
2
>>> np.linalg.matrix_rank(F)
2
>>> np.linalg.matrix_rank(G)
1
>>> np.linalg.matrix_rank(H)
1

```

Figure 9: Finding the rank of a matrix

dimension, vector space and spanned, may not be familiar to you unless you remember your college level linear algebra. If you fully understand what is meant by the rank of a matrix, I am happy and no further explanation is required. However, if you are not familiar

with the terms, think of it as the amount of information contained in the matrix. Again, this is a gross oversimplification and I am not proud of it. Figure 9 shows how the ranks of matrices are found out. Matrix A has rank 3 because none of its rows can be obtained from any other rows in it. Matrix B's rank is just 1, because the second and third rows can be obtained by multiplying the first row ([1, 2, 3]) with 2 and 3, respectively. Matrix C only has one non-zero row and hence has rank 1. Similarly, the ranks of the other matrices can also be understood. Rank is very relevant in our discussion and we will revisit this topic in later articles in this series.

Now it is time for us to wind up our discussion for this month. In the next article in this series,

we will add more tools to our arsenal so that they can be used for developing AI and machine learning based applications. We will also discuss terms like neural networks, supervised learning, unsupervised learning, etc, in more detail. Further, from the next article onwards, we will use JupyterLab in place of the Linux terminal. **END** 🐧

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